Zachary Kuptsow

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SUMMARY

Full-stack engineer who owns the product surfaces and the infrastructure they run on, with 5+ years across fintech, logistics, and AI products. Most recently founding lead engineer for an AI-powered mortgage platform, end to end: backend, frontend, infrastructure, CI/CD, security and compliance. Lays the groundwork so each new feature starts out as dependable as the one before it.

WORK EXPERIENCE

Low Rate Co. (Remote)

Founding Lead Engineer *March 2026 - September 2026*

- Set architecture and technical direction for an AI-powered mortgage lead-conversion platform managing a $60M loan pipeline, built on Python/Quart, GCP/Terraform, and React/TypeScript.
- Roughly 2,100 failed AI agent tasks had accumulated undetected for three weeks across three production queues; found the hourly health probe that counted only pending work and had reported healthy for 21 straight days, then closed the monitoring gap behind it.
- Led detection, containment, and recovery from a targeted supply-chain attack as sole responder, then shipped fail-closed CI scanning, immutable dependency pinning, and hardened org-level controls that automatically blocked subsequent attempts.
- Automated loan-document generation and delivery: Fannie Mae 1003/URLA PDFs and MISMO 3.4 XML conforming to the external industry schema, and idempotent outbound sync to the loan origination system behind transactional debounce locking, removing manual application re-keying.
- Architected a reusable async-retry framework powering all production LLM workloads (Cloud Tasks-backed deferred queues, dead-letter handling, and operator kill switches), plus a code-generation scaffold so every new AI feature inherits the same reliability guarantees by default.
- Built the release platform on GitHub Actions: gated PRs, automated Terraform and container rollouts, keyless authentication with no long-lived credentials in CI, and verified promotion from staging to production.
- Reduced Cloud Run spend by 76% and removed daily latency spikes through metrics-driven right-sizing and a scheduled pre-warm/scale-to-zero pattern.
- Eliminated production 504s and cut Firestore read operations with composite indexes, report date filters pushed into the query layer, and parallelized aggregation queries.
- Redesigned the customer dashboard in seven flag-gated phases with no disruption to live users.

DrayNow (Conshohocken, PA)

Software Engineer *July 2024 - March 2026*

- Drove backend development across a 13-service distributed system, serverless and event-driven, using Node.js, TypeScript, and PostgreSQL on AWS (Lambda, Step Functions, SQS/SNS, S3, ECS) managed through CDK, for an intermodal freight marketplace that moves tens of thousands of loads per month.
- Authentication migration, as technical lead: replaced a 6-year-old homegrown service with Kinde to achieve SOC2 compliance, a 4-month effort touching 400,000+ lines of code across two mobile apps, the web frontend, the public API, and internal services, delivered with zero downtime behind feature flags and backward-compatible tokens.

- Reduced AWS database costs by ~10% and improved API latency by 30-50% by optimizing query patterns during the authentication migration.
- Upgraded Node.js from v16 to v22 across the core platform and more than 20 major dependencies, absorbing breaking changes across six major versions with zero downtime and ~15% better performance.
- Automated multi-tenant provisioning with AWS CDK build configurations, so adding a tenant across every environment took one config entry and no manual certificate or DNS work.
- Established CI/CD on GitHub Actions with automated coverage reporting, test sharding, and shared linting standards across backend services.
- Stood up DataDog dashboards and Slack-routed error alerting for on-call triage, set the platform-wide structured logging standard, and carried the weekly on-call rotation.

Teampay (Remote - New York, NY)

Software Engineer *September 2021 - February 2024*

- Wrote and maintained RESTful APIs for a corporate spend platform processing roughly $2B in monthly transaction volume, using Python, Django, and PostgreSQL, with React and TypeScript on the front end.
- Spend-approval rules engine, the platform's original core logic: rebuilt it from 1:1 manager-employee approvals to many-to-many group approvals, decoupling the relationship from the database layer upward while preserving backward compatibility.
- Extended the Forest Admin integration so Customer Success could safely modify production data through engineer-defined validation, cutting engineering escalations from 24% to 7%.
- Shipped SAML/SCIM SSO with Okta and OneLogin for enterprise onboarding, along with a self-service vendor portal and multi-line card logic for expense tracking.
- Resolved hundreds of escalated production issues using Bash and the Django shell in Dockerized environments, and managed weekly releases through GitHub Actions.

EDUCATION

University of Delaware

Master of Science - Cybersecurity
August 2019 - June 2021
Concentration: Secure Software

Bachelor of Electrical Engineering
August 2015 - June 2019
Minor: Integrated Design

SKILLS

Languages & Frameworks

- Python (Django, Quart/Flask)
- TypeScript / JavaScript (Node/Express)
- React / Tailwind CSS
- SQL / PostgreSQL
- Bash / shell scripting
- Terraform (HCL)
- Jest / Vitest / Pytest

Infrastructure & Tooling

- GCP (Cloud Run, Cloud Tasks, Firestore, Pub/Sub, BigQuery)
- AWS (Lambda, Step Functions, SQS/SNS, ECS, S3, CDK)
- GitHub Actions
- Docker
- Supabase / Firebase
- LaunchDarkly

Practices & Systems

- AI/LLM engineering (OpenAI, Anthropic, agentic pipelines)
- Supply-chain security & incident response
- Event-driven architecture
- Observability & monitoring (DataDog)
- Performance & cloud cost optimization